

Terminal SoC Weight — Impact Analysis

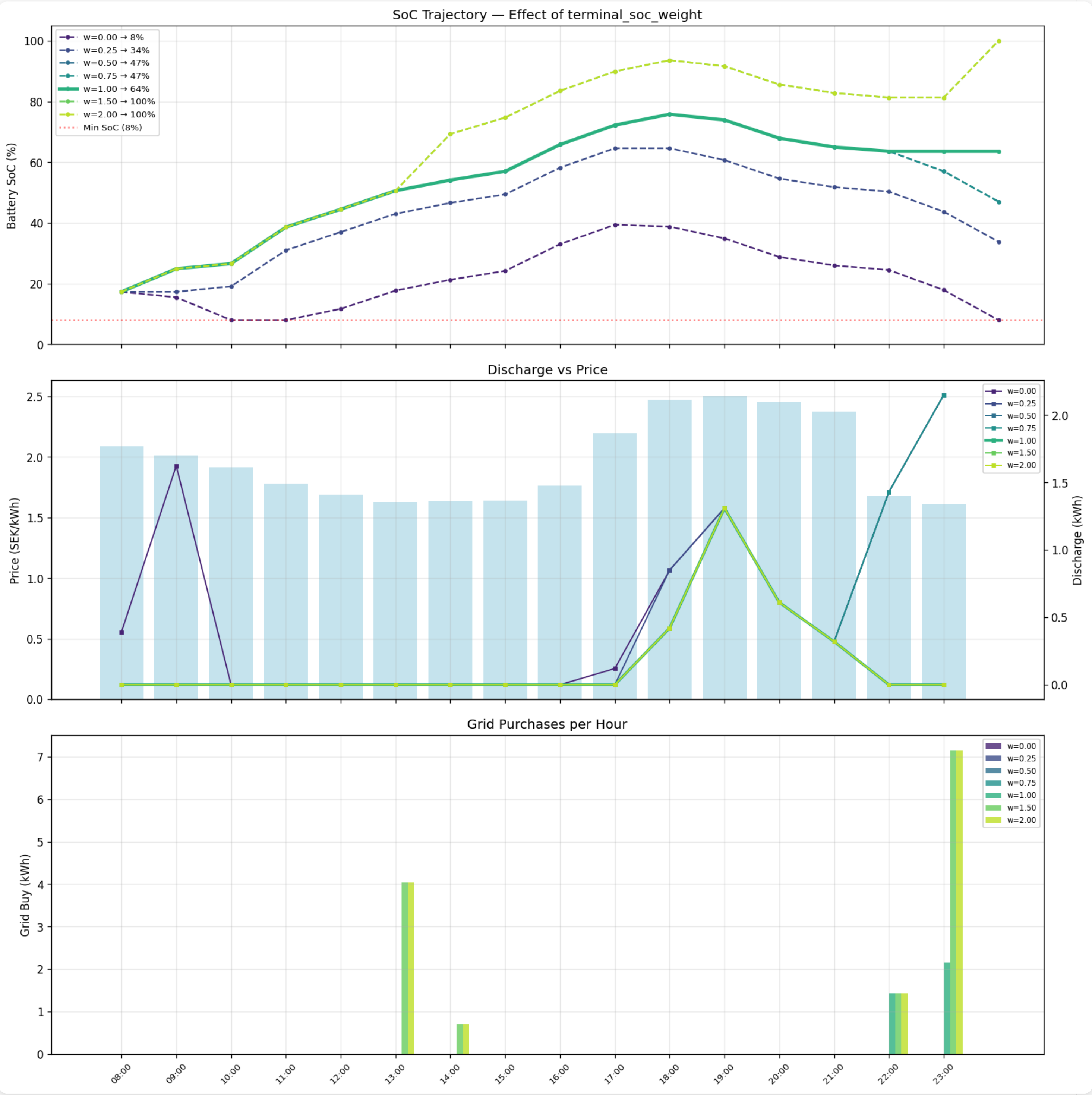
Real data from HA #1 (April 19, hours 08–23). Battery: 24.06 kWh, $\eta=90\%/90\%$, $\text{SoC}_0=17.3\%$.

Summary Table

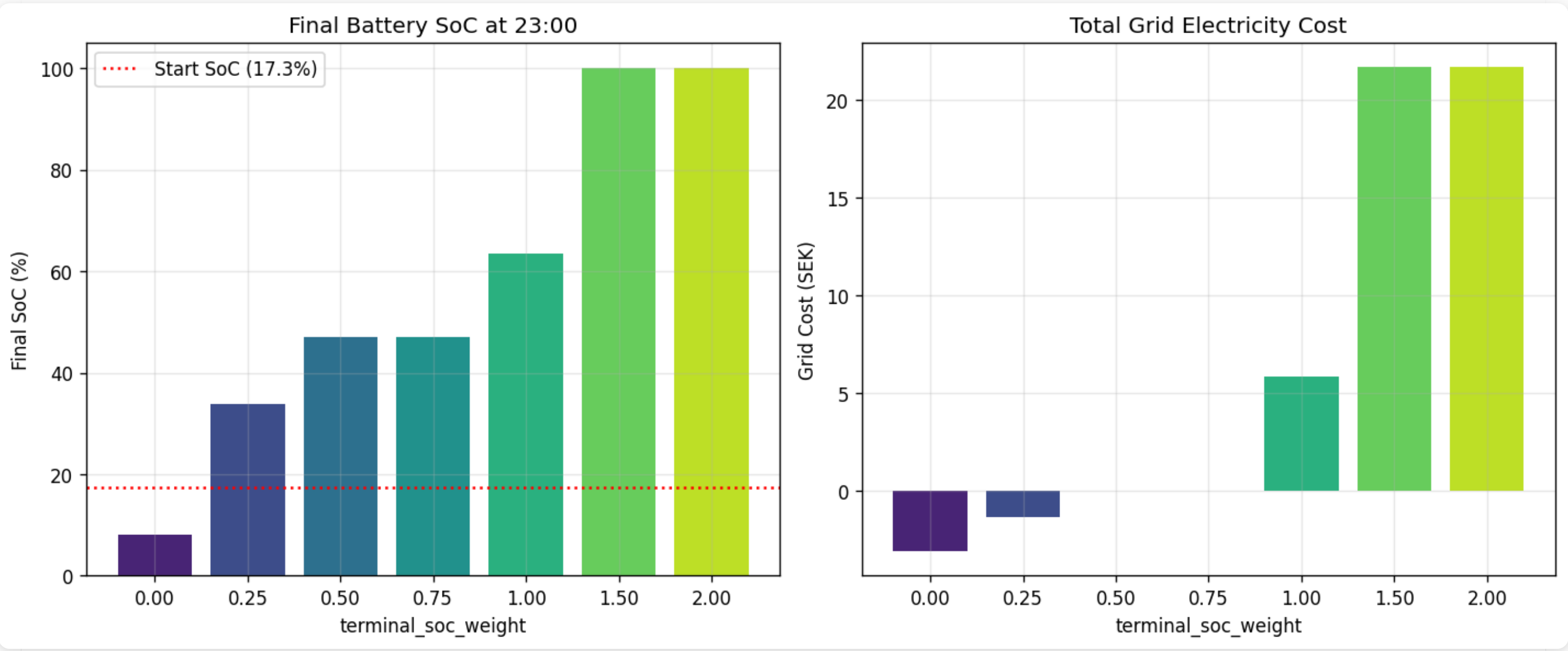
Weight	Final SoC	Grid Buy (kWh)	Grid Sell (kWh)	Grid Cost (SEK)	Discharged (kWh)
0.00	8.0%	0.00	9.83	-3.08	8.80
0.25	33.8%	0.00	3.43	-1.32	6.67
0.50	47.0%	0.00	0.00	0.00	6.24
0.75	47.0%	0.00	0.00	0.00	6.24
1.00	63.6%	3.58	0.00	5.87	2.66
1.50	100.0%	13.32	0.00	21.67	2.66
2.00	100.0%	13.32	0.00	21.67	2.66

Current setting: **w=1.00** (highlighted in green). Lower weight → more aggressive discharge at peak, less energy held overnight.

SoC Trajectory / Discharge / Grid Buy



Final SoC & Grid Cost Comparison



Key Observations

- w=0.00:** Battery fully empties — maximises today's savings but leaves nothing for tomorrow.
- w=0.50:** Good compromise — discharges more at evening peak while retaining some overnight energy.
- w=1.00 (current):** Conservative — holds ~64% overnight, buys from grid at 22–23h despite having stored energy.
- w=2.00:** Extremely conservative — battery barely discharges at all.